

41891/42891 Cloud Computing Infrastructures

Spring 2026

Assignment 1

Due date: 17:00 Friday, 28 August 2026.

Weighting: 20%

Assignment Objectives

1. Compare and contrast the components of Cloud computing infrastructure differentiating from the key stakeholders perspectives.
2. Hypothesize the business case for virtualisation and cloud computing given the businesses requirements and strategic needs.
3. Summarise the current theory behind Infrastructure as a Service (IaaS) platforms and be able to analyse alternatives.
4. Compare and contrast the core architectures and technologies used within current Cloud Platforms.

Assignment Requirements

This assignment is to be done **individually**, where you are required to discuss eight small topics on the use of cloud within businesses and virtualisation technologies. You must answer each question with either a short descriptive answer or with a diagram.

You are expected to do your own private research for this assignment. The material in the lecture notes alone might not intended to be sufficient. Your assignment must list all references used in an accepted academic referencing style (Harvard referencing preferred, but other styles permitted), and you should use in-line citations to reference sources.

Questions

Question 1 (5 Points): What is cloud (2 Points)? What is cloud computing (1 Point)? Explain the cloud cost model (2 Points).

Question 2 (15 Points): What are the cloud service models? (3 Points) Describe what are IaaS, PaaS and SaaS and the relationships among them. (12 Points)

Question 3 (15 Points): What are the cloud components? (5 Points) Please describe the cloud architecture based on these components. (10 Points)

Question 4 (20 Points): What is virtualisation (5 Points)? Describe different layers of virtualisation (5 Points). What is hypervisor (5 Points)? What does a hypervisor do (5 Points)?

Question 5 (15 Points): What is storage virtualisation (2 Points)? What does storage virtualisation do (5 Points)? When should storage virtualisation be used (3 Points)? What is SAN (2 Points)? What does SAN do (3 Points)?

Question 6 (10 Points): Explain at least five challenges existing in Building an IaaS Platform and how you can deal with them.

Question 7 (10 Points): What is Network as a Service (5 points)? Explain the cloud ecosystems (5 Points).

Question 8 (10 Points): What is a data center (5 Points)? List some basic components of a data center (5 Points).

Submission

Please submit Assignment 1 by 5:00pm Friday, 28 August 2026. The softcopy in doc or pdf needs to be submitted via Canvas. **No hardcopy** is needed. **Do NOT email** your report.

A late penalty of 20% working per day will apply for late submissions; if more than seven days late, the student will receive zero marks. Special considerations for late submission must be arranged by Week 4 with the Subject Coordinator. Formal documentation must be provided.

Marking Criteria for report

Specific marks for each sub-question are shown in the question list provided earlier. In questions where a justification is required, at least half of the marks available for that question will be allocated to the justification.

Academic Standards

Students are reminded of the principles laid down in the Faculty's Statement of Academic Integrity

- Good Practice and Ethics in Informal Assessment found at;
<wiki.it.uts.edu.au/start/Academic_Integrity>.

The University's rules regarding academic misconduct can be found at:
<<http://www.gsu.uts.edu.au/rules/student/section-16.html> >

Assignments in this Subject should be your own original work. The inclusion in assessable work of any material such as code, graphics or essay text obtained from other persons or sources without citation of the source is plagiarism and is a breach of University Rule 16.

Asking, or paying, anyone else to write any part of your assignments is considered cheating. This especially includes using outsourcing websites. If you are unsure if your activities (or other student's activities) will be considered cheating, ask your tutor or lecturer for advice.

Copying the work of other students currently or formerly enrolled in the subject is totally unacceptable behaviour.

All text written in assignments must be your own words, except for short, quoted, and clearly referenced sections. Text copied from web pages, articles or other sources, and not referenced, will be viewed as plagiarism and forwarded to the Faculty Conduct Committee as misconduct. Referencing styles may be found at the BELL web site <www.bell.uts.edu.au/referencing>.

Any collaboration with another person should be limited to those described in the "Acceptable Behaviour" section of the Statement of Academic Integrity. Any infringement by a student will be considered a breach of discipline and will be dealt with in accordance with the Rules and By-Laws of the University. Students are not to give to or receive from any other persons' copies of their assessable work in any form (hard copy or an electronic file). To do so is 'academic misconduct' and is a breach of University Rule 16. That is, assisting other students to cheat or to act dishonestly in a submitted assignment.

Accidental submission of another student's work as your own is considered to be a breach of University Rule 16 in that you are acting dishonestly since you should not have a copy of another student's work.